

ANUDIP FOUNDATION — MCA PLACEMENT TEAM

Project Documentation: Business & Functional Requirements (BRD / FRD)

Project: MCA Digital Attendance Portal

Team: Yellow Team

Organization: Anudip Foundation

Version: 1.0

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PART 1: BUSINESS REQUIREMENTS DOCUMENT (BRD)

1. Executive Summary

The purpose of this project is to centralize, digitize, and automate daily attendance tracking for student batches under the Anudip Foundation MCA curriculum. The application provides real-time tracking, database persistence, and automated executive email reporting to streamline placement and academic compliance monitoring.

2. Business Objectives & Vision

- Eliminate manual paper-based or spreadsheet attendance tracking.
- Provide instant metric summaries (Total Enrolled, Present Count, Absent Count, and Percentage Attendance).
- Automated submission of daily status reports to program stakeholders via formatted HTML email.
- Standardize attendance records across 30+ students per batch for placement eligibility evaluation.

3. Key Stakeholders

- Project Sponsor:** Anudip Foundation Placement Team
- Primary Recipient:** Placement Coordinator (ajay.sawarnkar@anudip.org)
- System End-Users:** Batch Trainers & Student Leads (Yellow Team)

4. Project Scope

In-Scope: Web-based Dashboard (Bootstrap 5 UI), Student Enrollment/Roster Management, One-Click Attendance Status Toggling, Real-Time KPI Analytics, and Automated SMTP Email Reporting Engine.

Out-of-Scope (Future): Biometric/RFID Scanner Integration, Facial Recognition-based Auto-Attendance, Student Self-Service Portal.

PART 2: FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

1. Technical Architecture & Stack

- Frontend:** HTML5, CSS3, Bootstrap 5.3, FontAwesome 6, JavaScript (Fetch API)
- Backend:** Python 3 (Flask Framework)
- Database:** SQLite3 (anudip_attendance.db)
- Email Engine:** Python smtplib (SSL Port 465) + email.mime

2. Database Schema Specification

Table	Column Name	Data Type	Key / Constraint
students	id	INTEGER	PRIMARY KEY AUTOINCREMENT
students	student_id	TEXT	UNIQUE, NOT NULL

students	name	TEXT	NOT NULL
attendance	id	INTEGER	PRIMARY KEY AUTOINCREMENT
attendance	student_id	TEXT	FOREIGN KEY, NOT NULL
attendance	date	TEXT	NOT NULL (YYYY-MM-DD)
attendance	status	TEXT	'Present' / 'Absent'

3. Core Functional Modules

- **FR-1 Roster Management:** Allows registering new students with validation against duplicate Roll IDs.
- **FR-2 Date-based Retrieval:** Enables switching dates and dynamically querying student records via SQL LEFT JOIN.
- **FR-3 One-Click Status Toggle:** Uses asynchronous AJAX POST calls to update SQLite via UPSERT statements.
- **FR-4 Real-Time Analytics:** Calculates Total Enrolled, Present, Absent, and Attendance Percentage on the fly.
- **FR-5 Executive Email Dispatcher:** Generates styled HTML emails containing summary cards and color-coded roster tables.